

$$\frac{1}{2} \lim_{\substack{\ell \rightarrow \{ \vec{w} \in V^2 \mid \|\vec{w}\| = 1 \} \\ \vec{w} \perp \{a, b\} \subseteq \ell}} \|\vec{w}\| \in \ell$$

parametric equation: $\begin{cases} x = x_0 + a \cdot t \\ y = y_0 + b \cdot t \end{cases}$

Cartesian Equations

a) Symmetric form: $\frac{x-x_0}{a} = \frac{y-y_0}{b}$

$$x = x_0, \text{ if } a = 0$$

$$y = y_0, \text{ if } b = 0$$

if $a, b \neq 0$, implicit form: $b(x-x_0) - a(y-y_0) = 0$

explicit form: $y = \frac{b}{a}x + y_0 - \frac{b}{a}x_0$, $\frac{b}{a}$ slope

The direction vectors of ℓ are $\vec{d} = \frac{1}{\sqrt{a^2+b^2}} \begin{pmatrix} b \\ -a \end{pmatrix}$

normal vectors of ℓ are $\vec{n} = \frac{1}{\sqrt{a^2+b^2}} \begin{pmatrix} a \\ b \end{pmatrix}$ $\vec{d} \perp \vec{n} \Rightarrow \langle \vec{d}, \vec{n} \rangle = \langle \frac{1}{\sqrt{a^2+b^2}} \begin{pmatrix} b \\ -a \end{pmatrix}, \frac{1}{\sqrt{a^2+b^2}} \begin{pmatrix} a \\ b \end{pmatrix} \rangle = \frac{1}{a^2+b^2} \langle \begin{pmatrix} b \\ -a \end{pmatrix}, \begin{pmatrix} a \\ b \end{pmatrix} \rangle = \frac{1}{a^2+b^2} (ba - ab) = 0$

1. Determine parametric equations for the line ℓ in the following cases:

- ℓ contains the point $A(1,2)$ and is parallel to the vector $a(3,-1)$,
 - ℓ contains the origin and is parallel to $b(4,5)$,
 - ℓ contains the point $M(1,7)$ and is parallel to Oy ,
 - ℓ contains the points $M(2,4)$ and $N(2,-5)$.
2. For the lines ℓ in the previous exercise
- give a Cartesian equation for ℓ ,
 - describe all direction vectors for ℓ .
3. Determine a Cartesian equations for the line ℓ in the following cases:
- ℓ has slope -5 and contains the point $A(1,-2)$,
 - ℓ has slope 1 and is at distance 2 from the origin,
 - ℓ contains the point $A(-2,3)$ and has an angle of 60° with the Ox -axis,
 - ℓ contains the point $B(1,7)$ and is orthogonal to $\vec{n}(4,3)$.
4. For the lines ℓ in the previous exercise
- give parametric equations for ℓ ,
 - describe all normal vectors for ℓ .

1,2) a) $b(x-x_0) - a(y-y_0) = 0$
 $-1(x-1) - 3(y-2) = 0$
 $-x+1-3y+6=0$
 $-x-3y+7=0$
 $\vec{d} = \langle 3, -1 \rangle$
 $\vec{n} = \langle -1, -3 \rangle$

b) $O(0,0) \in \ell, \ell \parallel \vec{b}(4,5)$

Parametric equations

$$\begin{cases} x = x_0 + a \cdot t \\ y = y_0 + b \cdot t \end{cases} \Rightarrow \begin{cases} x = 4t \\ y = 5t \end{cases}$$

Cartesian form: $\frac{x-x_0}{a} = \frac{y-y_0}{b} \rightarrow$ Symmetric Form

$$\frac{x}{4} = \frac{y}{5}$$

Implicit form: $b(x-x_0) - a(y-y_0) = 0$

$$5x - 4y = 0$$

Explicit form: $y = \frac{b}{a}x + y_0 - \frac{b}{a}x_0$

$$y = \frac{5}{4}x$$

Normal vectors of ℓ are $\vec{n} = \langle 5, -4 \rangle$

Dir vectors of ℓ are $\vec{d} = \langle 4, 5 \rangle$

c) $M(1,7) \in \ell$
 $\ell \parallel Oy \Rightarrow a = 0$
Parametric eq: $\begin{cases} x = 1 \\ y = 7 + bt, b \in \mathbb{R} \end{cases}$

Cartesian eq: $x = 1$

Normal vectors: $\vec{n} = \langle b, 0 \rangle = \langle 1, 0 \rangle$

Dir vectors: $\vec{d} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

d) $M(2,4), N(2,-5) \in \ell$

$$\vec{MN} = (0, -9)$$

Parametric eq: $\begin{cases} x = 2 \\ y = 4 - 9t \end{cases}$

$$\vec{d} = \langle 0, 1 \rangle$$

$$\vec{n} = \langle -1, 0 \rangle$$

3,4) a) $\frac{y-y_0}{b} = -5 = m = \frac{y_1-y_0}{x_1-x_0}$

$$y - y_0 = -5(x - x_0)$$

$$y + 2 = -5(x - 1)$$

$$y + 2 = -5x + 5$$

$$5x + y - 3 = 0$$

Parametric: $\begin{cases} x = t \\ y = 3 - 5t \end{cases}$

$$\vec{d} = \langle 1, -5 \rangle$$

$$\vec{n} = \langle 5, 1 \rangle$$

b)

c) $A(1,-2,3)$

$$\neq \langle \vec{d}, \vec{0} \rangle = 60^\circ \Rightarrow m(\vec{d}) = \frac{1}{2} \left(\frac{\pi}{3} \right) = \sqrt{3}$$

$$\frac{b}{a} = \sqrt{3} \Rightarrow b = a\sqrt{3} \Rightarrow \vec{d} = \langle 1, \sqrt{3} \rangle$$

$$\vec{d} = \langle \sqrt{3}, 1 \rangle$$

$$\frac{y-y_0}{b} = \frac{x-x_0}{a} \Leftrightarrow \frac{y-3}{a\sqrt{3}} = \frac{x-2}{a} \Leftrightarrow \frac{y-3}{\sqrt{3}} = x-2 \Rightarrow x = \frac{y-3}{\sqrt{3}} + 2$$

$$y - 3 = (x+2)\sqrt{3}$$

$$y - x\sqrt{3} = 2\sqrt{3} + 3$$

$$\begin{cases} x = t \\ y = 2\sqrt{3} + 3 + t\sqrt{3} \end{cases}$$

d) $\vec{v} \perp \vec{n} \Rightarrow \vec{v} \cdot \vec{n} = 0$

$$4a = -3b$$

$$\frac{b}{a} = -\frac{4}{3} \Rightarrow \vec{d} = \langle -\frac{3}{4}, 1 \rangle$$

$$\vec{d} = \langle 1, \frac{3}{4} \rangle$$

$$\frac{y-2}{\frac{3}{4}} = \frac{x-1}{1}$$

$$3y - 2 = 4x - 4$$

$$4x - 3y + 2 = 0$$

$$\begin{cases} x = t \\ y = \frac{4t+2}{3} \end{cases}$$

$$\vec{d} = \langle 1, \frac{4}{3} \rangle$$

$$\vec{n} = \langle 4, -3 \rangle$$

b)

5. Consider a line ℓ . Show that

c) if $\vec{v}(v_1, v_2)$ is a direction vector for ℓ then $\vec{n}(v_2, -v_1)$ is a normal vector for ℓ ,

d) if $\vec{n}(n_1, n_2)$ is a normal vector for ℓ then $\vec{v}(n_2, -n_1)$ is a direction vector for ℓ .

a) $\vec{v}(v_1, v_2)$ b) $\vec{n}(n_1, n_2)$

$$\vec{v} \perp \vec{n} \Leftrightarrow \vec{v} \cdot \vec{n} = 0 \quad \vec{n} \perp \vec{v} \Leftrightarrow \vec{n} \cdot \vec{v} = 0$$

$$v_1 n_1 + v_2 n_2 = 0 \quad n_1 v_1 + n_2 v_2 = 0$$

6. Consider the points $A(1,2)$, $B(-2,3)$ and $C(4,7)$. Determine the medians of the triangle ABC .

$A(x_A, y_A)$ $B(x_B, y_B)$ $C(x_C, y_C)$
 $x_A = \frac{x_B + x_C}{2} = 1$ $y_A = \frac{y_B + y_C}{2} = 5$
 $x_B = \frac{x_A + x_C}{2} = 2.5$ $y_B = \frac{y_A + y_C}{2} = 4.5$
 $x_C = \frac{x_A + x_B}{2} = 2.5$ $y_C = \frac{y_A + y_B}{2} = 3.5$

Parametric eq: $\begin{cases} x = \frac{5}{2} + \frac{3}{2}t \\ y = \frac{9}{2} + \frac{3}{2}t \end{cases}$

c) $A(x_A, y_A)$ $B(x_B, y_B)$ $C(x_C, y_C)$
 $x_A = \frac{x_B + x_C}{2} = -\frac{1}{2}$ $y_A = \frac{y_B + y_C}{2} = \frac{5}{2}$
 $x_B = \frac{x_A + x_C}{2} = \frac{5}{2}$ $y_B = \frac{y_A + y_C}{2} = \frac{9}{2}$
 $x_C = (-\frac{1}{2}, \frac{5}{2})$

Parametric eq: $\begin{cases} x = -\frac{1}{2} + \frac{3}{2}t \\ y = \frac{5}{2} + \frac{3}{2}t \end{cases}$

$$\vec{d} = \langle 1, 1 \rangle$$

$$\vec{n} = \langle 1, -1 \rangle$$

8. Let $A(1,3)$, $B(-4,3)$ and $C(2,9)$ be the vertices of a triangle. Determine

a) the length of the altitude from A .

b) the line containing the altitude from A .

$$A_{ABC} = \frac{1}{2} \begin{vmatrix} 1 & 3 & 1 \\ -4 & 3 & 1 \\ 2 & 9 & 1 \end{vmatrix} = \frac{1}{2} \cdot 30 = 15$$

$$A_{ABC} = \frac{d(A, BC) \cdot BC}{2} = 15 \Rightarrow d(A, BC) = \frac{\sqrt{2}}{2}$$

$$BC = \sqrt{6^2 + 6^2} = 6\sqrt{2}$$

b) $m_{BC} = \frac{y_C - y_B}{x_C - x_B} = \frac{9-3}{2-(-4)} = \frac{6}{6} = 1 \Rightarrow m_{AA'} = -1$

$$y - y_A = m_{AA'}(x - x_A)$$

$$y - 3 = -1(x - 1)$$

$$3x + y - 4 = 0$$

$$y = 4 - 3x$$

9. Determine the circumcenter of the triangle with vertices $A(1,2)$, $B(3,-2)$, $C(5,6)$.

$M_{BC} = \frac{x_C - x_B}{y_C - y_B} = \frac{2}{8} = \frac{1}{4}$
 $OA \perp BC \Rightarrow m_{OA} = -4$

$$m_{AC} = \frac{y_C - y_A}{x_C - x_A} = \frac{4}{4} = 1 \quad A'(x_A', y_A') = (\frac{3}{2}, \frac{1}{2}) \quad C'(\frac{3}{2}, \frac{1}{2})$$

$$m_{AB} = -1 \quad B'(4, 2) \quad C'(2, 0)$$

$$OA': y = -4x + y_0 + 4x_0 \quad OC': y = -x + y_0 + x_0$$

$$y = -4x + 2 + 4 = 10 - 4x \quad y = -x + 2 + 2 = -x + 4$$

$$4x + y = 10 \quad x + y = 4$$

$$\begin{cases} 4x + y = 10 \\ x + y = 4 \end{cases} \Rightarrow \begin{cases} x = \frac{16}{3} \\ y = -\frac{10}{3} \end{cases}$$

10. Determine the angle between the lines $\ell_1: y = 2x + 1$ and $\ell_2: y = -x + 2$.

$$\ell_1: y = 2x + 1 \Rightarrow \frac{y}{x} = 2 \Rightarrow b_1 = 2 \Rightarrow \ell_1(\ell_1, \ell_2)$$

$$\ell_2: y = -x + 2 \Rightarrow \frac{y}{x} = -1 \Rightarrow \ell_2(\ell_1, \ell_2)$$

$$\vec{v}_1 \cdot \vec{v}_2 = \|\vec{v}_1\| \|\vec{v}_2\| \cos(\theta)$$

$$\vec{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \vec{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \Rightarrow \cos(\theta) = \frac{1}{\sqrt{5} \sqrt{2}} = \frac{\sqrt{10}}{10}$$

$$\theta = \arccos\left(\frac{\sqrt{10}}{10}\right)$$

11. Let $A(1,-2)$, $B(5,4)$ and $C(-2,0)$ be the vertices of a triangle. Determine the equations of the angle bisectors for the angle $\angle A$.

$$\frac{AB}{AC} = \frac{BP}{PC} \quad AB = \sqrt{20} = 2\sqrt{5} \quad AC = \sqrt{13}$$

$$\Rightarrow P\left(\frac{x_B + x_C}{3}, \frac{y_B + y_C}{3}\right) = P\left(2, \frac{4}{3}\right)$$

$$\vec{AP} = \begin{pmatrix} 1 \\ \frac{10}{3} \end{pmatrix}$$

$$A(1, -2)$$

$$\begin{cases} x = 1 + 2t \\ y = -2 + \frac{10}{3}t \end{cases}$$

$$x - 1 = \frac{y + 2}{5} \Rightarrow 5x - y - 7 = 0$$

12. Let A' be the orthogonal reflection of $A(10,10)$ in the line $\ell: 3x + 4y - 20 = 0$. Determine the coordinates of A' .

$$\ell: 3x + 4y - 20 = 0$$

$$4y = -3x + 20 \Rightarrow y = -\frac{3}{4}x + 5$$

$$y = -\frac{3}{4}x + 5$$

$$\vec{n} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

$$m_{AA'} = -\frac{4}{3} \Rightarrow \vec{AA'} = \langle 4, -3 \rangle$$

$$\frac{x_0 - x_A}{3} = \frac{y_0 - y_A}{4}$$

$$4x_0 - 3y_0 = 3y_0 - 4x_0$$

$$\begin{cases} 4x_0 - 3y_0 = 10 \\ 3x_0 + 4y_0 = 20 \end{cases} \Rightarrow \begin{cases} x_0 = 4 \\ y_0 = 2 \end{cases}$$

$$\begin{cases} x_0 = \frac{x_A + x_{A'}}{2} \\ y_0 = \frac{y_A + y_{A'}}{2} \end{cases} \Rightarrow \begin{cases} x_{A'} = -2 \\ y_{A'} = -6 \end{cases}$$

13. Determine Cartesian equations for the lines passing through $A(-2,5)$ which intersect the coordinate axes in congruent segments.

Let a be the distance

I $M(a, a)$ II $M(a, -a)$

$$\vec{AM} = (-2, a-5) \quad \vec{AM} = (-2, -a-5)$$

$$\frac{x_M + 2}{-2} = \frac{y_M - 5}{a-5} \quad \frac{x_M + 2}{-2} = \frac{y_M - 5}{-a-5}$$

$$\frac{a+2}{-2} = \frac{-5}{-a-5} \quad \frac{-a+2}{-2} = \frac{-5}{-a-5}$$

$$(a+2)(a-5) = 10 \quad (a-2)(a-5) = -10$$

$$a^2 - 3a - 10 = 10 \quad a^2 - 7a + 10 = 0$$

$$a^2 - 3a - 20 = 0 \quad a = 4 \text{ or } 10$$

$$a = 3 + 8 = 11$$

$$a_1 = \frac{3 + \sqrt{89}}{2}$$

$$a_2 = \frac{3 - \sqrt{89}}{2}$$

14. Determine Cartesian equations for the lines situated at distance 4 from the line $12x - 5y - 15 = 0$.

Since the distance is 4, $\ell \parallel \ell_1$

$$d(\ell, \ell_1) = d(P, \ell_1)$$

$$P(x_0, y_0)$$

$$d(P, \ell_1) = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}} \Rightarrow \frac{|12x_0 - 5y_0 - 15|}{\sqrt{13}} = 4$$

$$|12x_0 - 5y_0 - 15| = 52 \Rightarrow \begin{cases} 12x_0 - 5y_0 - 67 = 0 \\ 12x_0 - 5y_0 + 37 = 0 \end{cases}$$

15. Determine the values k for which the distance from the point $(2,3)$ to the line $8x + 15y + k = 0$ equals 5.

$$d(P, \ell) = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$$

$$\frac{|8 \cdot 2 + 15 \cdot 3 + k|}{\sqrt{8^2 + 15^2}} = 5$$

$$|151 + k| = 85 \Rightarrow k \in \{30, -132\}$$

16. Consider the points $A(3,-1)$, $B(9,1)$ and $C(-5,5)$. For each pair of these three points, determine the line which is equidistant from them.

Let $\ell_i: a_i x + b_i y + c_i = 0, d(A, \ell_i) = d(B, \ell_i)$

$$\ell_1: a_1 x + b_1 y + c_1 = 0, d(A, \ell_1) = d(C, \ell_1)$$

$$\ell_2: a_2 x + b_2 y + c_2 = 0, d(B, \ell_2) = d(C, \ell_2)$$

$$d(A, \ell_1) = \frac{|a_1 x_A + b_1 y_A + c_1|}{\sqrt{a_1^2 + b_1^2}} = \frac{|a_1 x_B + b_1 y_B + c_1|}{\sqrt{a_1^2 + b_1^2}}$$

$$d(B, \ell_1) = \frac{|a_1 x_B + b_1 y_B + c_1|}{\sqrt{a_1^2 + b_1^2}} = \frac{|a_1 x_C + b_1 y_C + c_1|}{\sqrt{a_1^2 + b_1^2}}$$

$$\Rightarrow |a_1 x_A + b_1 y_A + c_1| = |a_1 x_B + b_1 y_B + c_1|$$

17. The point $A(3,-2)$ is the vertex of a square and $M(1,1)$ is the intersection point of its diagonals. Determine Cartesian equations for the sides of the square.

$$\vec{AC} = 2\vec{AM} = 2\vec{BM} = 2\vec{DM}$$

$$\vec{AC} = 2\vec{AM} = 2\vec{BM} = 2\vec{DM} \Rightarrow \begin{cases} C(-1, 4) \\ B(3, 4) \\ D(-1, -2) \end{cases}$$

$$AB: x = 3$$

$$BC: y = -2$$

$$CD: x = -1$$

$$DA: y = 4$$

18. Determine a point on the line $5x - 4y - 4 = 0$ which is equidistant to the points $A(1,0)$ and $B(-2,1)$.

$$P \in \ell: 5x_0 - 4y_0 - 4 = 0$$

$$AP = BP$$

$$AP = \sqrt{(x_0 - 1)^2 + y_0^2} \quad BP = \sqrt{(x_0 + 2)^2 + (y_0 - 1)^2} \Rightarrow (x_0 - 1)^2 + y_0^2 = (x_0 + 2)^2 + (y_0 - 1)^2$$

$$2x_0 - 1 = 2(2x_0 + 1)$$

$$2x_0 - 1 = 6x_0 + 3$$

$$4x_0 = 3x_0 + 2 \Rightarrow x_0 = 2$$

$$4x_0 = 5x_0 - 4 \Rightarrow x_0 = 4$$

$$7x_0 = 12 \Rightarrow x_0 = \frac{12}{7}$$

$$y_0 = 2 - \frac{36}{7} = -\frac{22}{7}$$

19. The point $A(2,0)$ is the vertex of an equilateral triangle. The side opposite to A lies on the line $x + y - 1 = 0$. Determine Cartesian equations for the lines containing the other two sides.

Let $AB = AC = BC$

$$\begin{cases} (x_B - 2)^2 + y_B^2 = (x_C - 2)^2 + y_C^2 \\ x_B^2 + y_B^2 = x_C^2 + y_C^2 \end{cases}$$

$$\begin{cases} x_B + y_B - 1 = 0 \\ x_C + y_C - 1 = 0 \end{cases}$$